# Two-Stage Neural Reranking for Document Retrieval: Bridging Lexical and Semantic Relevance

*Text Retrieval and Search Engines – Final Project Part B*

## 1. Model Description and Experimental Setting

**The Vocabulary Mismatch Problem:** Classical retrieval models like BM25 and Query Likelihood rely on exact lexical matching—they cannot identify relevance when query and document use different words for the same concept. A query for "automobile safety" will miss documents about "car crash prevention" despite semantic equivalence. Our innovation addresses this gap through neural reranking that captures deep semantic relationships.

**Our Innovation – Two-Stage Neural Architecture:** We propose a retrieve-then-rerank pipeline combining the *efficiency* of statistical retrieval with *semantic understanding* of transformers.

**Stage 1 (QLD)**: Query Likelihood with Dirichlet smoothing retrieves 1,000 candidates in milliseconds using

$$P(w|d) = (tf + \mu \cdot P(w|C)) / (|d| + \mu).$$

**Stage 2 (Cross-Encoder)**: BGE reranker (BAAI/bge-reranker-base) reranks top-100, applying deep semantic analysis where it matters most—at positions that dominate MAP.

**Why Cross-Encoders Excel:** The key innovation is *joint query-document encoding*. Unlike bi-encoders that encode separately, cross-encoders concatenate: *[CLS] query [SEP] document [SEP]*, enabling **full cross-attention**—every query token attends to every document token across 12 transformer layers. The model captures fine-grained interactions: "FDA" relates to "Food and Drug Administration," "polio vaccine" connects to "immunization programs." The [CLS] token aggregates these into relevance score:

$$score(q,d) = \sigma(W \cdot h[CLS] + b).$$

**Model Selection:** BGE-reranker-base was trained on MS MARCO (500K+ query-passage pairs) with contrastive loss, achieving SOTA on the BEIR benchmark. Crucially, it requires *no task-specific fine-tuning*—the pre-trained model generalizes directly to ROBUST04's news domain, avoiding overfitting risks inherent when training on only 50 queries.

**Method Selection – Why Neural Reranking Over LambdaMART:** We also implemented LambdaMART, a gradient-boosted learning-to-rank model that achieved higher MAP on the *training set* (50 queries). However, on the held-out *test set* (199 queries), **the neural reranker outperformed LambdaMART**. This demonstrates a critical insight: LambdaMART, trained on limited queries, overfitted to training patterns, while the pre-trained cross-encoder - never exposed to ROBUST04 during training-generalized better to unseen queries. This validates our design principle: *leveraging large-scale pre-training (MS MARCO's 500K examples) transfers more effectively than task-specific training on small datasets.*

**Experimental Setup:** ROBUST04 collection (528,030 documents), 50 train / 199 test queries. QLD μ=1000 (tuned). Reranking top-100 candidates, max 512 tokens. Platform: Databricks with GPU (~0.3s/query for 100 candidates). Pipeline preserves QLD ranking for positions 101-1000.

## 2. Related Work

**[1] Nogueira & Cho (2019)** pioneered BERT passage reranking (+27% MRR on MS MARCO), establishing the retrieve-then-rerank paradigm we adopt. **[2] Xiao et al. (2023)** introduced BGE rerankers with contrastive learning, achieving SOTA on BEIR—we leverage their pre-trained model. **[3] Zhai & Lafferty (2004)** established Dirichlet smoothing for language model IR (our Stage 1). **[4] Burges (2010)** introduced LambdaMART for learning-to-rank; while powerful with sufficient training data, our results show pre-trained neural models generalize better in low-resource settings.

**Comparison:** Unlike dense retrievers (DPR, ColBERT) requiring pre-computed embeddings, our two-stage design uses standard inverted indices, making it immediately deployable. Unlike LambdaMART requiring labeled training data, the cross-encoder provides zero-shot transfer—a crucial advantage when relevance judgments are scarce.

## References

[1] Nogueira & Cho, "Passage Re-ranking with BERT," *arXiv:1901.04085*, 2019. [2] Xiao et al., "C-Pack: Packaged Resources for General Chinese Embedding," *arXiv:2309.07597*, 2023. [3] Zhai & Lafferty, "A Study of Smoothing Methods for LM Applied to IR," *ACM TOIS* 22(2), 2004. [4] Burges, "From RankNet to LambdaRank to LambdaMART," *MSR Tech Report*, 2010.